Nowcasting Illiquid NAVs with a State Space Model

Inspired by Brown, G. W., Ghysels, E., Gredil, O. (2022). Nowcasting net asset values. Journal of Finance

February 11, 2026

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Motivation

Private investments: characteristics and data challenges (1/3)

What distinguishes private investments

- Investments in **non-listed assets** such as private equity, private credit, real assets and infrastructure.
- Positions are valued through **periodic appraisals**, not continuous market prices.
- Capital is deployed and returned through **irregular cash flows** over long investment horizons.

Private investments: characteristics and data challenges (2/3)

How this is reflected in reported data

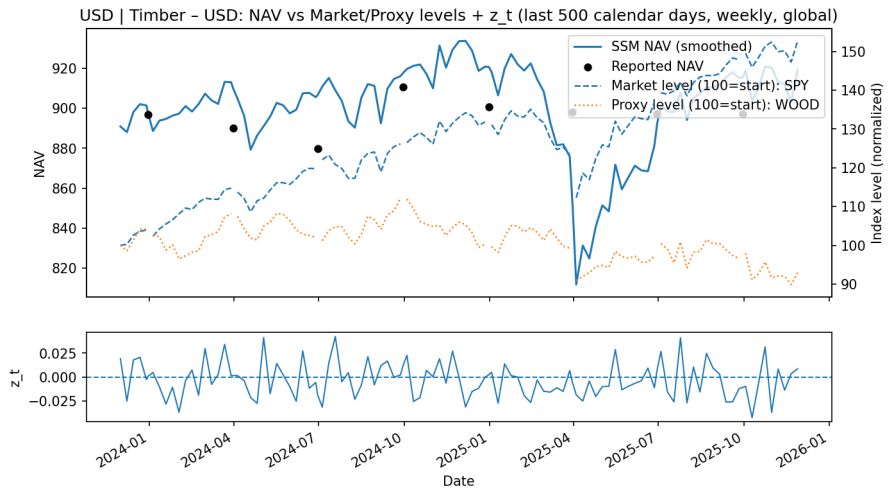
- Net asset values (NAVs) are reported **quarterly or monthly**, often with a delay.
- Reported NAVs embed **appraisal smoothing** and valuation conventions.
- Observed NAV changes combine **economic performance**, timing effects and **measurement noise**.

Private investments: characteristics and data challenges (3/3)

Resulting challenge for risk and portfolio analysis

- Raw NAV-based returns are not directly comparable to liquid asset returns.
- Volatility, correlation and persistence inferred from NAVs can be misleading.
- Additional structure is required to recover economically meaningful return dynamics.

Example: USD Timber (market and proxy levels + z_t)



Top panel: SSM NAV vs reported NAV with normalized market and proxy levels. Bottom panel: proxy residual z_t.

Kalman Filer and Brown, G. W., Ghysels, E., Gredil, O. (2022) (BBG) approach (1/3)

Kalman filtering: the estimation backbone

- Developed to estimate **unobservable states** from noisy and incomplete data.
- In a **linear Gaussian setting**, the Kalman filter and smoother provide the **optimal linear estimator** of the latent state.
- Naturally handles **missing observations**, **irregular timing**, and **multiple information sources**.
- Widely used in economics, finance, and engineering when the object of interest cannot be observed directly.

Kalman Filer and Brown, G. W., Ghysels, E., Gredil, O. (2022) (BBG) approach (2/3)

BBG approach: state of the art for private assets

- Builds on the Kalman framework to model **private asset returns and NAV reporting**.
- Explicitly separates **true economic returns** from **appraisal smoothing** and reporting noise.
- Uses **cumulative returns** and **smoothing dynamics** that match how NAVs are actually constructed in practice.
- Demonstrates **systematic improvements** over interpolation, AR-based unsmoothing, and proxy-only methods.

Kalman Filer and Brown, G. W., Ghysels, E., Gredil, O. (2022) (BBG) approach (3/3)

Why this matters for our application

- Private investment returns are **latent by construction**, even after smoothing.
- The Kalman–BBG framework extracts the **maximum information** from sparse NAVs, market data, and proxies in a coherent way.
- This makes it particularly well suited for producing **high-frequency, risk-consistent return series** for portfolio construction and risk management.

A state space model (SSM): challenges and solutions (1/3)

Private investment challenge	Tactical or simple approach and limitation	Strategic solution
Sparse and irregular NAV observations	**Interpolation** between NAV points introduces artificial smoothness and ignores **estimation uncertainty**.	**Latent weekly state** handles missing observations. **All available information** is used consistently as it arrives.
Appraisal smoothing embedded in reported NAVs	**Autoregressive or lag regressions** fit reported returns but do not model the **underlying smoothing process**.	**Explicit smoothing mechanism** separates **economic performance** from valuation and reporting effects.

A state space model (SSM): challenges and solutions (2/3)

Private investment challenge	Tactical or simple approach and limitation	Strategic solution
Distorted volatility and correlation from raw NAVs	Using **NAV or proxy returns directly** understates risk and can misrepresent **diversification benefits**.	**Recovery of latent returns** produces **economically meaningful volatility and correlation** for risk analysis.
Systematic and idiosyncratic return drivers are mixed	**Single regressions** often conflate market and idiosyncratic effects, leading to **unstable or misleading exposures**.	**State equation identifies market exposure** β, while **innovations capture idiosyncratic shocks**.

A state space model (SSM): challenges and solutions (3/3)

Private investment challenge	Tactical or simple approach and limitation	Strategic solution
Need for a coherent and scalable framework	**Multiple ad hoc models** require manual tuning, ongoing monitoring, and **subjective judgement**.	**Single estimation framework** with interpretable parameters delivers **consistent outputs across strategies**.

Model Inputs

Data universe and coverage

Scope of the investment universe for this study

- **Private assets (Burgiss / MSCI)**: around **90 strategy indexes** across private equity, private credit, real assets and infrastructure.
- **Market factors**: regional equity and credit indexes (US, Europe, UK, APAC).
- **Proxies**: liquid, well traded ETFs selected by region, asset class and strategy type.

Proxy selection reflects economic exposure and ensures liquidity and data availability.

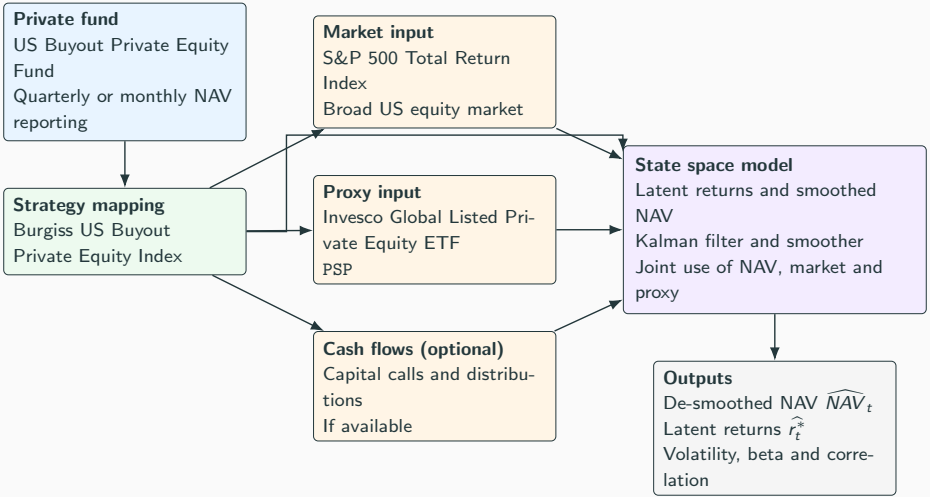
Time series availability

Data type	Frequency / history
Burgiss / MSCI private asset indexes	Quarterly, long history (some series from late 1970s)
Market indexes	Daily or weekly, multi-decade coverage
Proxy ETFs	Daily or weekly, post-ETF era

Illustrative mapping examples

Fund (example)	Private strategy	Market index	Proxy ETF
US Buyout PE Fund	Burgiss US Buyout PE	S&P 500 TR	Listed Private Equity ETF (PSP)
Senior Credit Fund	Burgiss Private Credit – Senior	Regional credit index	Senior loan ETF

Data Flow: an illustrative example of US Buyout Private Equity Fund



Model specification

Model idea: latent returns, smoothing, and optimal filtering

Latent economics and reporting mechanics (high level)

State (latent, weekly)

$$x_t = \begin{bmatrix} r_t^* \\ s_t \\ m_t \end{bmatrix} \qquad r_t^* = \alpha + \beta r_{m,t} + \eta_t$$

$$s_t = s_{t-1} + r_t^* \qquad m_t = (1-\lambda)s_t + \lambda m_{t-1}$$

Return construction

- r_t^* is the **true (unobserved) return**.
- s_t is the **cumulative true return** implied by r_t^*.
- m_t is a **smoothed cumulative return**, reflecting appraisal smoothing via λ.

Measurement (observed, sparse)

$$\log NAV_t^{rep} = m_t + \varepsilon_t^{nav} \quad \text{(only at reporting dates)}$$

$$z_t = \beta_c r_t^* + \varepsilon_t^{px} \quad \text{(proxy signal)}$$

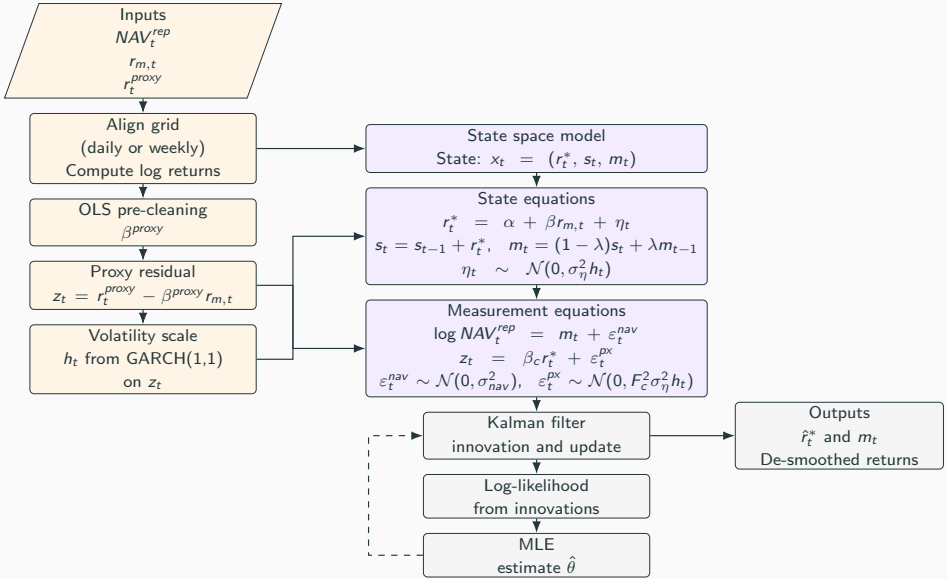
Key implications

- Reported NAVs are **sparse and noisy** observations of cumulative performance.
- Even the **smoothed component** m_t is **latent** and not directly observed.
- Information about returns arrives both through NAV reports and higher-frequency proxies.

Inference logic

- The true return process r_t^* is never observed directly.
- Reported NAVs provide low-frequency information about smoothed cumulative returns.
- The state space model combines these signals to infer the full latent path (r_t^*, s_t, m_t) between reports.
- In a linear Gaussian setting, the Kalman filter provide the **optimal linear estimator** of this latent state.

Algorithm flow: from input data to de-smoothed returns



Results

Volatility comparison: Tactical vs Strategic (Weekly SSM) Approach

Panel averages (where Tactical is available, n=67):

Tactical = 2.8% Weekly SSM = 13.0% Market = 13.8% Proxy = 14.3% Δ = **+10.2 pp**

Name	Tactical	Weekly SSM	Market	Proxy	Δ (pp)
AUD — Debt – Senior – AUD	4.0%	3.3%	6.4%	2.8%	-0.7
EUR — Debt – Mezzanine – EUR	3.3%	4.7%	5.3%	4.5%	1.3
GBP — Real Estate – Ex-Value-Added – APAC – GBP	1.9%	8.8%	16.7%	10.9%	6.9
CHF — Real Estate – Ex-Value-Added – UK – CHF	1.7%	9.6%	12.6%	16.9%	7.9
USD — Real Estate – Ex-Value-Added – Europe – USD	2.5%	13.1%	15.0%	13.2%	10.6
GBP — Real Estate – Ex-Value-Added – Europe – GBP	1.3%	13.1%	15.0%	13.2%	11.8
CHF — Real Estate – All – USA – CHF	2.6%	15.0%	15.1%	15.4%	12.4
EUR — Venture Capital – EUR	4.0%	22.7%	15.0%	34.4%	18.7
AVG (n=67)	**2.8%**	**13.0%**	**13.8%**	**14.3%**	**10.2**

Δ is Weekly SSM minus Tactical (percentage points). Averages are computed over the n=67 subset where Tactical is available.

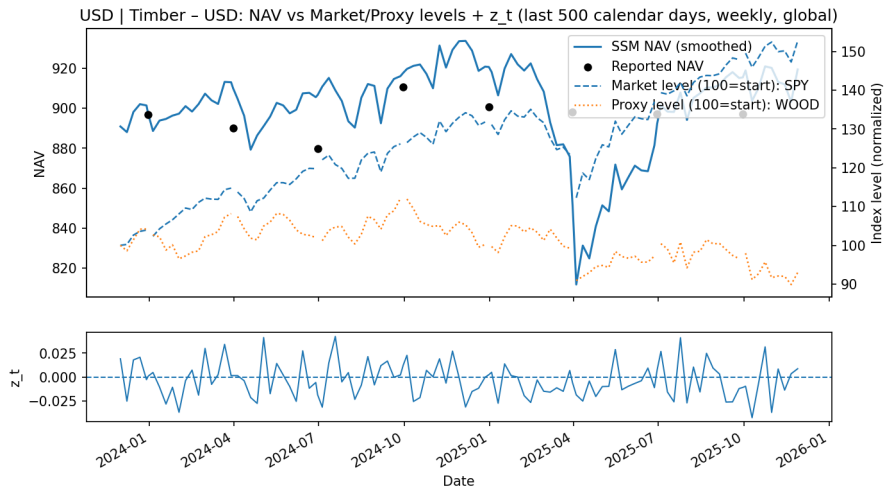
Correlation and persistence summary (n=90)

Panel averages (all funds, n=90): NAV–Market = 0.80 NAV–Proxy = 0.88 Market–Proxy = 0.58 ACF(1) = -0.00

asset_id	corr_nav_market	corr_nav_proxy	corr_market_proxy	acf1
SEK — Real Estate – Ex-Value-Added – UK – SEK	0.48	0.98	0.49	-0.24
AUD — Real Estate – Value-Added – Europe – AUD	0.64	1.00	0.64	0.06
USD — Timber – USD	0.72	0.91	0.61	-0.30
AUD — Infrastructure – AUD	0.85	0.91	0.61	-0.00
USD — Real Estate – Value-Added – USA – USD	0.91	0.84	0.54	-0.09
CHF — Debt – Mezzanine – CHF	0.95	0.71	0.49	0.06
USD — Real Estate – Value-Added – UK – USD	1.00	0.74	0.67	0.10
AVG (n=90)	**0.80**	**0.88**	**0.58**	**-0.00**

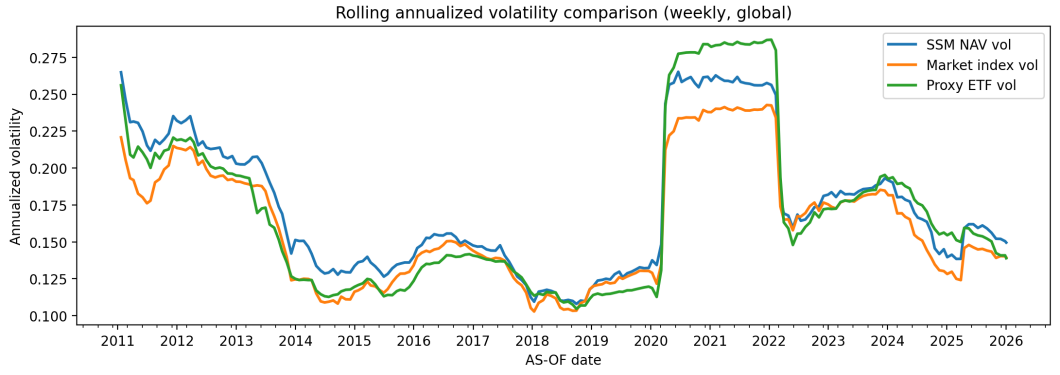
Rows shown are representative quantiles of `corr_nav_market` across the full panel. ACF(1) refers to autocorrelation of SSM-implied returns.

Example: USD Timber (market and proxy levels + z_t)



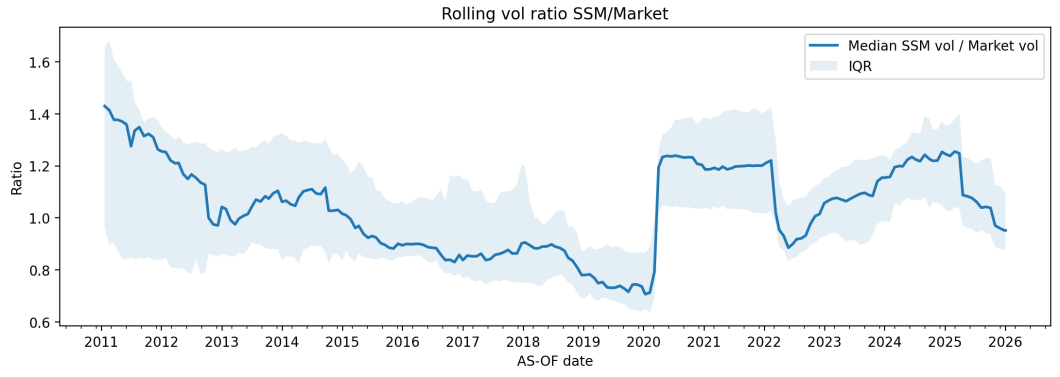
Top panel: SSM NAV vs reported NAV with normalized market and proxy levels. Bottom panel: proxy residual z_t.

Rolling Annualized Volatility: SSM vs Market vs Proxy



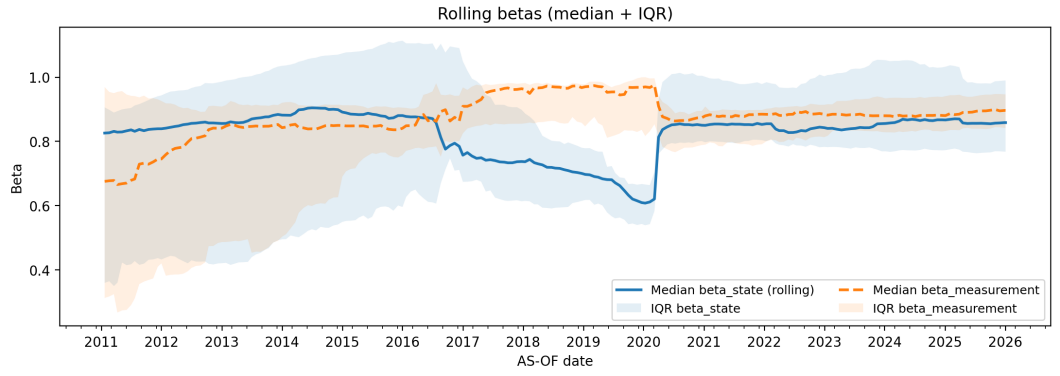
- Reported NAV volatility is often lower than market and proxy volatility.
- The state space model (SSM) produces higher, economically plausible volatility.
- Volatility spikes coincide with major market stress episodes (GFC, COVID).
- Appraisal smoothing dampens reported NAV fluctuations.
- The SSM extracts latent return dynamics consistent with market regimes.
- Results align with the core findings in the NAV de-smoothing literature.

Volatility Uplift: SSM Relative to Market



- Median ratio of SSM volatility to market volatility is above one.
- The shaded band represents the interquartile range (IQR) across funds.
- Dispersion increases during stressed periods.
- On average, de-smoothed risk exceeds reported market index risk.
- IQR highlights cross-sectional heterogeneity while remaining robust to outliers.
- This provides a stable, interpretable measure of volatility correction.

Rolling Market Betas: State and Measurement Layers



- Median market beta from the state equation remains stable over time.
- The interquartile range captures cross-sectional dispersion.
- Measurement beta remains small, reflecting controlled proxy noise.
- Latent returns exhibit economically meaningful market exposure.
- Stability supports the identification of systematic risk rather than noise.
- This is consistent with the literature's emphasis on interpretable betas.

Appendix

State Space Model: States and Dynamics

State vector

$$x_t = \begin{bmatrix} r_t^* \\ s_t \\ m_t \end{bmatrix}$$

State equations

$$r_t^* = \alpha + \beta r_{m,t} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2 h_t)$$

$$s_t = s_{t-1} + r_t^*$$

$$m_t = (1-\lambda)\, s_t + \lambda\, m_{t-1}$$

Variance factor

$$h_t \text{ from GARCH(1,1)}$$

Compact matrix form

$$x_t = A x_{t-1} + B_t + \Gamma\, \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2 h_t)$$

State components

- r_t^*: latent unsmoothed return (daily or weekly).
- s_t: cumulative latent log return.
- m_t: smoothed latent level driving reported NAV.

Drivers

- $r_{m,t}$: market factor return (broad index).
- α: drift of latent return r_t^*.
- β: loading on market factor in return dynamics.
- ψ: AR(1) persistence of r_t^*.

Smoothing

- λ: appraisal/smoothing parameter.
- m_t moves slowly when λ is close to 1 (very sticky NAV).

Volatility

- σ_η: scale of latent shocks η_t.
- h_t: time-varying volatility factor shared with proxy noise.

State Space Model: Measurements and Observation Vector

Inputs from data

- Reported NAVs: NAV_t^{rep}.
- Market returns: $r_{m,t}$. Proxy returns: r_t^{proxy}.

Step 1: proxy residual (pre-processing)

$$z_t \equiv r_t^{proxy} - \beta^{proxy} r_{m,t},$$

Step 2: measurement equations

$$\log NAV_t^{rep} = m_t + \varepsilon_t^{nav}$$

$$z_t = \beta_c r_t^* + \varepsilon_t^{px}$$

Noise distributions

$$\varepsilon_t^{nav} \sim \mathcal{N}(0, \sigma_{nav}^2), \qquad \varepsilon_t^{px} \sim \mathcal{N}(0, F_c^2 \sigma_\eta^2 h_t)$$

Observation vector for Kalman filter

$$y_t = \begin{bmatrix} \log NAV_t^{rep} \\ z_t \end{bmatrix} = \begin{bmatrix} m_t \\ \beta_c r_t^* \end{bmatrix} + \begin{bmatrix} \varepsilon_t^{nav} \\ \varepsilon_t^{px} \end{bmatrix}$$

Measurement roles

- NAV equation links m_t (smoothed level) to reported NAVs.
- Proxy equation links r_t^* (latent return) to proxy residual z_t.
- Together they pin down both the level and dynamics of r_t^*.

Parameters in measurement layer

- β_c: sensitivity of proxy residuals z_t to latent return r_t^*.
- σ_{nav}: cross-sectional noise in reported NAVs.
- F_c: relative noise scale of proxy vs. latent process.

Data → model → KF

1. From raw data compute $r_{m,t}$, r_t^{proxy}, NAV_t^{rep}.
2. Estimate β^{proxy}; build residual z_t.
3. Run GARCH(1,1) on proxy residuals $\Rightarrow h_t$.
4. At each t, feed $y_t = (\log NAV_t^{rep}, z_t)$ into KF.
5. The filter updates beliefs about $x_t = (r_t^*, s_t, m_t)$.

Kalman Filter and Maximum Likelihood

Parameters

$$\theta = (\alpha, \beta, \lambda, \psi, \beta_c, \sigma_\eta^2, \sigma_{nav}^2, F_c)$$

Measurement vector

$$y_t = \begin{bmatrix} \log NAV_t^{rep} \\ z_t \end{bmatrix}$$

Kalman filter (schematic)

$$\begin{aligned}
\hat{x}_{t|t-1} &= A\hat{x}_{t-1|t-1} + B_t && \text{(predict state)} \\
P_{t|t-1} &= AP_{t-1|t-1}A^\top + Q && \text{(predict covariance)} \\
v_t &= y_t - H\hat{x}_{t|t-1} && \text{(innovation)} \\
\Omega_t &= HP_{t|t-1}H^\top + R && \text{(innov. covariance)} \\
K_t &= P_{t|t-1}H^\top\Omega_t^{-1} && \text{(Kalman gain)} \\
\hat{x}_{t|t} &= \hat{x}_{t|t-1} + K_t v_t && \text{(update state)}
\end{aligned}$$

Log-likelihood

$$\log L(\theta) = -\tfrac{1}{2}\sum\left(\log|\Omega_t| + v_t^\top\Omega_t^{-1}v_t\right)$$

Notation

- x_t: latent state (r_t^*, s_t, m_t).
- y_t: stacked observations $(\log NAV_t^{rep}, z_t)$.
- A, Q: state transition matrix and noise covariance matrix.
- H, R: observation matrices and measurement matrix covariances.
- $P_{t|t-1}$: covariance of prediction.
- $P_{t|t}$: covariance of updated state.
- v_t, Ω_t: innovation vector and its variance.

Estimation steps

1. Choose candidate θ within bounds.
2. Run Kalman filter, compute v_t and Ω_t.
3. Build $\log L(\theta)$ from innovations.
4. Maximize $\log L(\theta)$.
5. Run smoother at $\hat{\theta}$ to recover full latent path.

State equation in matrix form

State vector

$$x_t = \begin{bmatrix} r_t^* \\ s_t \\ m_t \end{bmatrix}$$

Scalar shock and variance

$$\eta_t \sim \mathcal{N}(0, \sigma_\eta^2 h_t), \qquad Q_t = \sigma_\eta^2 h_t$$

State-space form

$$x_t = Ax_{t-1} + B_t + \Gamma \eta_t$$

Matrices

$$A = \begin{bmatrix} \psi & 0 & 0 \\ \psi & 1 & 0 \\ (1-\lambda)\psi & (1-\lambda) & \lambda \end{bmatrix}, \qquad B_t = \begin{bmatrix} \alpha + \beta r_{m,t} \\ \alpha + \beta r_{m,t} \\ (1-\lambda)(\alpha + \beta r_{m,t}) \end{bmatrix}, \qquad \Gamma = \begin{bmatrix} 1 \\ 1 \\ 1-\lambda \end{bmatrix}$$

Here h_t is a pre-computed volatility factor (for example from GARCH(1,1) on proxy residuals) used to scale the shock variance.

Measurement equation and KF objects

Observation vector

$$y_t = \begin{bmatrix} \log NAV_t^{rep} \\ z_t \end{bmatrix}$$

Measurement form

$$y_t = Hx_t + \varepsilon_t, \qquad \varepsilon_t = \begin{bmatrix} \varepsilon_t^{nav} \\ \varepsilon_t^{px} \end{bmatrix}, \qquad \varepsilon_t \sim \mathcal{N}(0, R_t)$$

Measurement matrix

$$H = \begin{bmatrix} 0 & 0 & 1 \\ \beta_c & 0 & 0 \end{bmatrix}$$

Observation noise covariance

$$R_t = \begin{bmatrix} \sigma_{nav}^2 & 0 \\ 0 & F_c^2 \sigma_\eta^2 h_t \end{bmatrix}$$

Kalman filter definitions used in the main slide

$$v_t = y_t - H\hat{x}_{t|t-1}, \qquad \Omega_t = HP_{t|t-1}H^\top + R_t, \qquad K_t = P_{t|t-1}H^\top \Omega_t^{-1}$$

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t v_t, \qquad P_{t|t} = (I - K_t H)P_{t|t-1}$$

Incorporating Cash Flows (Optional Extension)

Motivation

Reported NAV changes reflect both

- economic performance of the underlying assets, and
- investor cash flows such as capital calls and distributions.

How cash flows enter the model (paper suggestion)

Cash flows are treated as **observed inputs** (actual realized fund cash flows), not as latent processes. A common way is to adjust the NAV observation with a deterministic cash-flow term:

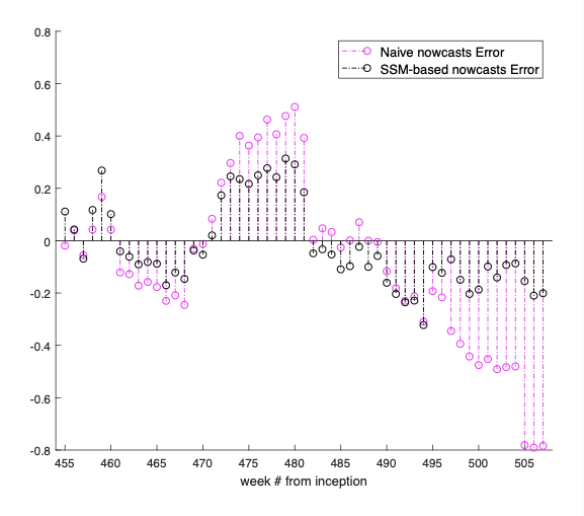
$$\log NAV_t^{rep} = m_t + c_t + \varepsilon_t^{nav}, \qquad c_t = \log\left(1 + \frac{CF_t}{NAV_{t-1}}\right)$$

- CF_t aggregates net cash flows over the period (calls minus distributions, plus fees if modeled).
- c_t shifts the observed NAV mechanically, so the latent state focuses on economic performance.
- The latent return dynamics r_t^* are unchanged; only the NAV measurement equation is augmented.

Interpretation

- Latent returns r_t^* represent the **pure performance** component.
- Cash flows explain part of NAV moves that are not performance.
- This can reduce bias when flows are large or irregular.

SSM reduces nowcasting errors (synthetic experiment)



Simulated fund: naive vs SSM nowcasting errors (true NAVs are known in simulation only).

Purpose of the experiment

- Conducted on **synthetic data**.
- True NAVs and true returns are known.
- Allows clean comparison of methods.

Exercise steps

1. Simulate latent fund returns:

$$r_t^* = \beta r_{m,t} + \eta_t$$

2. Construct true asset values by cumulating r_t^*.
3. Generate reported NAVs via smoothing:

$$\bar{r}_{0:t} = (1-\lambda) r_{0:t} + \lambda \bar{r}_{0:t-1}$$

   plus reporting noise.
4. Compute naive nowcasts using market or proxy returns.
5. Estimate SSM and recover r_t^*.

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